

PREDICTIVE MAINTENANCE OF VEHICLES USING MACHINE LEARNING

RESEARCH GAP IDENTIFICATION:

1. Limited Availability of Public Real-World Datasets

Most existing studies rely on proprietary, simulated, or domain-specific datasets that are not publicly available. This limits the reproducibility of research and makes it difficult to compare the performance of different predictive maintenance models under similar conditions. The lack of standardized logistics fleet datasets also affects the practical applicability of existing solutions.

2. Insufficient Feature Leakage Analysis

Many studies focus on improving prediction accuracy without examining whether certain input features unintentionally contain information related to the target variable. The inclusion of leakage-prone features can produce unrealistically high prediction accuracy while reducing the model's reliability in real-world deployment.

3. Limited Comparison of Class Imbalance Handling Techniques

Although predictive maintenance datasets often suffer from class imbalance, only a limited number of studies investigate how balancing techniques influence model performance. Comprehensive comparisons between models trained on the original dataset and those trained using balancing methods are still lacking.

4. Limited Evaluation of Multiple Machine Learning Algorithms

Several studies focus on a single algorithm or a limited number of machine learning models. As a result, there is insufficient comparative analysis to identify the most suitable algorithm for predictive maintenance under different operational conditions and dataset characteristics.

5. Lack of Explainability in Predictive Models

Many predictive maintenance models prioritize prediction accuracy but provide limited explanation of how individual features influence maintenance decisions. This reduces user confidence and makes it difficult for maintenance engineers to interpret and validate the model's predictions.

6. Limited Focus on Logistics Fleet Vehicle Maintenance

A significant portion of existing research focuses on manufacturing equipment, industrial machinery, defence vehicles, or individual automotive components. Comparatively fewer studies investigate predictive maintenance specifically for commercial logistics fleet vehicles operating under diverse environmental and operational conditions.

7. Inadequate Integration of Environmental and Operational Factors

Most predictive maintenance models emphasize sensor readings and historical maintenance records while giving limited attention to external factors such as weather conditions, road conditions, route characteristics, and vehicle usage patterns. These factors can significantly influence vehicle health and maintenance requirements.

8. Limited End-to-End Predictive Maintenance Frameworks

Existing studies often focus on individual stages such as data preprocessing, model development, or evaluation. Few provide a complete framework that includes exploratory data analysis, feature leakage analysis, feature selection, preprocessing, model comparison, and comprehensive performance evaluation.

9. Limited Real-World Deployment Considerations

Most studies conclude after evaluating model performance and do not discuss practical deployment aspects such as integration with fleet management systems, model updating, scalability, or decision support for maintenance scheduling.